REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

High Performance Fibres

A full-screen, syllabus-style digital handbook for Course Code 6062B. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6062B

COURSE TITLE
High Performance Fibres

DEPARTMENT
Textile Technology

TYPE
Theory

SEMESTER
6

CATEGORY
Open Elective

USE
Study + notes PDF

FORMAT
Big handbook

6062B

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

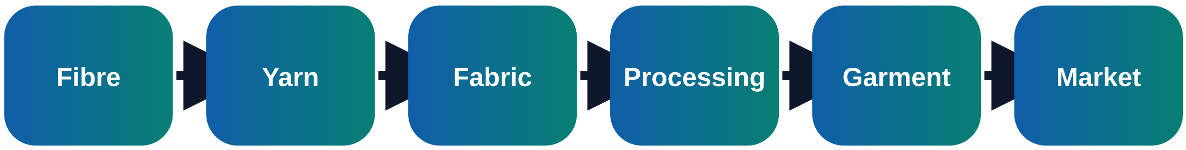
Course objectives and syllabus map

Objectives

- ✓ Classify high performance fibres and their properties
- ✓ Explain aramid, carbon, glass, UHMWPE and ceramic fibres
- ✓ Select fibres for protection, composites, filtration and industrial textiles

□□□□□□: □ □□□□ □□□□□□□□□□□□□ □□□□□□ □□□□. Textile industry-□ real work □□□□□□□□□□□ □□□□□□□□□□□□□□ calculation, quality check, documentation, reporting □□□□□□□□□ □□□□□□□□□□□□□ □□□.

Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Introduction and property requirements	Difference between conventional and high performance fibres; Key properties: tenacity, modulus, heat resistance, chemical resistance and flame resistance	Short notes, calculations, diagram, checklist, viva answers
M2	Organic high performance fibres	Aramid fibres: para-aramid, meta-aramid, structure and applications; UHMWPE fibres: high strength, low density and creep limitation	Short notes, calculations, diagram, checklist, viva answers
M3	Inorganic and carbon fibres	Glass fibre types and reinforcement applications; Carbon fibre manufacture, PAN precursor, stabilization and carbonization	Short notes, calculations, diagram, checklist, viva answers
M4	Testing, processing and applications	Testing of tensile strength, modulus, LOI, thermal stability and chemical resistance; Yarn and fabric formation problems due to stiffness and abrasion	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Introduction and property requirements

Core explanation

- ✓ Difference between conventional and high performance fibres
- ✓ Key properties: tenacity, modulus, heat resistance, chemical resistance and flame resistance
- ✓ Specific strength, creep, fatigue, abrasion and dimensional stability
- ✓ Applications in protective, aerospace, marine, automotive and civil engineering textiles

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□: Practical answer □□□□□□□□□□ definition □□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Organic high performance fibres

Core explanation

- ✓ Aramid fibres: para-aramid, meta-aramid, structure and applications
- ✓ UHMWPE fibres: high strength, low density and creep limitation
- ✓ PBO and high temperature organic fibres
- ✓ Safety, handling and cutting of high strength fibres

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□□: Practical answer □□□□□□□□□□□ definition □□□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 3

Inorganic and carbon fibres

Core explanation

- ✓ Glass fibre types and reinforcement applications
- ✓ Carbon fibre manufacture, PAN precursor, stabilization and carbonization
- ✓ Ceramic, basalt and metal fibres
- ✓ Comparison of properties and limitations

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□□: Practical answer □□□□□□□□□□□ definition □□□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Testing, processing and applications

Core explanation

- ✓ Testing of tensile strength, modulus, LOI, thermal stability and chemical resistance
- ✓ Yarn and fabric formation problems due to stiffness and abrasion
- ✓ Composite reinforcement, ballistic protection, ropes, filters and heat shields
- ✓ Recycling, sustainability and cost considerations

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□□: Practical answer □□□□□□□□□□□ definition □□□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Specific Strength = Tensile Strength / Density

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Modulus = Stress / Strain

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

LOI = Minimum oxygen concentration required to sustain burning

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Composite Strength depends on fibre strength, matrix and fibre orientation

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Compare aramid, carbon and glass fibre properties
- ✓ Select fibre for firefighter suit and justify
- ✓ Prepare application chart for technical textiles
- ✓ Interpret tensile test data of high strength yarn

Important keywords

Aramid · Carbon fibre · Glass fibre · UHMWPE · LOI · Modulus · Composite · Technical textile

Short questions

Q1.1	Explain: Difference between conventional and high performance fibres	Answer with definition, process, application and one example.
Q1.2	Explain: Key properties: tenacity, modulus, heat resistance, chemical resistance and flame resistance	Answer with definition, process, application and one example.
Q1.3	Explain: Specific strength, creep, fatigue, abrasion and dimensional stability	Answer with definition, process, application and one example.
Q2.1	Explain: Aramid fibres: para-aramid, meta-aramid, structure and applications	Answer with definition, process, application and one example.
Q2.2	Explain: UHMWPE fibres: high strength, low density and creep limitation	Answer with definition, process, application and one example.
Q2.3	Explain: PBO and high temperature organic fibres	Answer with definition, process, application and one example.
Q3.1	Explain: Glass fibre types and reinforcement applications	Answer with definition, process, application and one example.
Q3.2	Explain: Carbon fibre manufacture, PAN precursor, stabilization and carbonization	Answer with definition, process, application and one example.

Q3.3	Explain: Ceramic, basalt and metal fibres	Answer with definition, process, application and one example.
Q4.1	Explain: Testing of tensile strength, modulus, LOI, thermal stability and chemical resistance	Answer with definition, process, application and one example.
Q4.2	Explain: Yarn and fabric formation problems due to stiffness and abrasion	Answer with definition, process, application and one example.
Q4.3	Explain: Composite reinforcement, ballistic protection, ropes, filters and heat shields	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to Aramid. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define High Performance Fibres.
2. List four important keywords: Aramid, Carbon fibre, Glass fibre, UHMWPE.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Introduction and property requirements.
7. Compare two important concepts from Organic high performance fibres.
8. Explain the practical importance of Inorganic and carbon fibres in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for High Performance Fibres in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of High Performance Fibres.
2. Use keywords: Aramid, Carbon fibre, Glass fibre, UHMWPE, LOI, Modulus, Composite, Technical textile.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Introduction and property requirements

- Difference between conventional and high performance fibres
- Key properties: tenacity, modulus, heat resistance, chemical resistance and flame resistance
- Specific strength, creep, fatigue, abrasion and dimensional stability
- Applications in protective, aerospace, marine, automotive and civil engineering textiles

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Organic high performance fibres

- Aramid fibres: para-aramid, meta-aramid, structure and applications
- UHMWPE fibres: high strength, low density and creep limitation
- PBO and high temperature organic fibres
- Safety, handling and cutting of high strength fibres

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Inorganic and carbon fibres

- Glass fibre types and reinforcement applications
- Carbon fibre manufacture, PAN precursor, stabilization and carbonization
- Ceramic, basalt and metal fibres
- Comparison of properties and limitations

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Testing, processing and applications

- Testing of tensile strength, modulus, LOI, thermal stability and chemical resistance
- Yarn and fabric formation problems due to stiffness and abrasion
- Composite reinforcement, ballistic protection, ropes, filters and heat shields
- Recycling, sustainability and cost considerations

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.